BIO304: 1st MidSem Exam/Feb23/25 marks

1. Which of the following element is least abundant in our body? (a) Carbon (b) Phosphorous (c) Oxygen (d) Hydrogen
2. If for a biochemical reaction, $\Delta H<0$ and $\Delta S>0$, then (a) reaction is spontaneous (b) $\Delta G=0$ (c) disorder in the system with decrease if the reaction proceeds.
3. Isoelectric point of a protein is defined as the pH: (a) at which net charge on the molecule is zero (b) at which all groups are protonated (c) at which all groups are unprotonated (d) at which each acidic group is protonated and each basic group is unprotonated.
4. Which is the major force that stabilizes the three-dimensional structure of globular proteins?
5. Which of the following amino acids have side chains negatively charged under physiological condition of pH ~7? (a) Asp (b) His (c) Trp (d) Cys.
6. If a polypeptide has ~400 amino acid residues, what is its approximate mass?
7. Which of the following amino acids is expected to be found inside the core of a water-soluble protein? (a) Arg (b) His (c) Asp (d) Ile
8. In the α-helix the hydrogen bonds are formed between the CO groups of the n^{th} residue and NH group of the ___th number of residue.
9. Which one of the following statement about protein secondary structure is correct? (a) α-helix is primarily stabilized by ionic interactions between the side chains of amino acids (b) β-sheets exists only in anti-parallel form (c) β-turns often contain proline (d) α–helix can be composed of more than one polypeptide chain.
10. A twenty residue peptide composed of L-Ala forms a right handed alpha helix in methanol. If every alternate L-Ala is replaced with D-Ala, the peptide will probably (a) form a right handed alpha helix (b) form a left handed alpha helix (c) form ten residue long right handed alpha helix followed by ten residue long left handed helix (d) not form any kind of helix.
11. Peptide bond has a backbone of atoms in which of the following sequences? (a) C-N-N-C (b) C-C-C-N (c) C-C-N-C (d) N-C-C-C.
12. A Ramachandran plot describes, for a particular amino acid, the sterically permitted angles for rotations about (a) Cα-Cβ bond (b) Cα-C bond (c) N-Cα bond (d) Cα-H bond. Which two options are correct?
13. Which of the following bonds in proteins has a partial double bond character? (a) Cα-C (b) Cα-N (c) C-N (d) C-O.
14. Peptide bond in protein is (a) planar, but rotates to three preferred dihedral angles (b) nonpolar, but rotates to two preferred dihedral angles (c) nonpolar, and fixed in trans configuration (d) planar, and usually in trans configuration.
15. An α-helix is made up of 78 amino acid residues. What will be the value of its axial length?
16. Formation of an α-helix results in a dipole along the helix, with a positive N-terminus and negative C-terminus. What is the most likely explanation for this? (a) Helix dipole is the sum of each peptide bond dipole moment (b) N-terminus of a protein contains positively charged amine group and C-terminus contains negatively charged carboxyl group (c) Peptide carbonyl oxygens that form hydrogen bond with amide hydrogens point toward the C-terminus (d) Side chain residues in alpha helices extend outward, perpendicular to the helix axis.
17. What is the major force that determines complementary base pairing interactions in DNA?
18. What is the major force that stabilizes DNA double helix?
19. Which of the following is not correct about the collagen structure? (a) rich in glycine (b) rich in proline (c) rich in hydroxyproline (d) it is an α-helix.
20. Which vitamin is essential in the context of collagen structure-function? Deficiency of this vitamin causes___.
21. Collagen is best described as: (a) an α-helical structural protein (b) a coiled-coil found in hair (c) a cross-linked globular protein (d) a triple helical fibrous protein.
22. What repetitive motif is responsible for the collagen structure? (a) Ala-X-Y (b) Cys-X-Y (c) Gly-X-Y (d) Pro-X-Y.
23. Which of the following is/are the feature(s) of α-keratin? (a) presence of –S-S- crosslink between adjacent a-helices (b) presence of intrachain H-bonding (c) presence of intrachain -S-S- crosslink.
24. Using circular dichroism one can obtain: (a) three dimensional structure of a protein (b) amino acid sequence of a protein (c) quaternary structure of a protein (d) relative proportions of α-helices β-sheets and loops in a protein.
25. A double stranded DNA has 30% Thymine. The percentage of Cytosine is:________________.